Set 8 practice: Vector functions, directional derivatives, tangent planes

Zill §9.1, 9.5, 9.6

Quiz prep. Closed book, about 10 minutes per problem. Problems 1 and 2 are vector-valued functions of one variable (limits, derivatives, integrals, tangent vectors, arc length), problems 3 and 4 are gradients and directional derivatives, problems 5 and 6 are tangent planes and normal lines, and problems 7 and 8 use the same ideas to find special points and the angle between two surfaces. No figures are given, so describe the geometry in words as you go. Every answer was recomputed independently and compared with the written result.

Key methods

Vector functions. For $\mathbf{r}(t) = \langle x(t),\, y(t),\, z(t)\rangle$, limits, derivatives and integrals act on each component. The derivative $\mathbf{r}'(t)$ is tangent to the curve, and

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}, \qquad s = \int_a^b \|\mathbf{r}'(t)\|\, dt$$

Gradient and directional derivative. $\nabla f = \langle f_x, f_y, f_z\rangle$. For a **unit** vector $\mathbf{u}$,

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

Normalize any given direction first. The largest rate of change at a point is $\|\nabla f\|$, in the direction of ∇f. The smallest is $-\|\nabla f\|$, in the direction of $-\nabla f$. The rate is zero along directions perpendicular to ∇f.

Level surface $F(x, y, z) = c$**.** ∇F at a point is normal to the surface there. Tangent plane at (x_0, y_0, z_0) and normal line:

$$F_x(x - x_0) + F_y(y - y_0) + F_z(z - z_0) = 0, \qquad x = x_0 + F_x t, \quad y = y_0 + F_y t, \quad z = z_0 + F_z t$$

All three partials are evaluated at the point. For $z = f(x, y)$ use $F = f(x, y) - z$, so the normal is $\langle f_x, f_y, -1\rangle$. The angle between two surfaces is the angle between their normals.

Each problem is on its own page with space for your working; the solution is on the next page.

Problem P1

(a) Evaluate $\lim_{t\to 0}\left\langle \dfrac{\sin 3t}{t},\ \dfrac{1-\cos t}{t^2},\ \dfrac{e^{2t}-1}{t}\right\rangle$.

(b) For $\mathbf{r}(t) = \langle te^t,\ \sin 2t,\ \dfrac{1}{1+t^2}\rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}''(0)$.

(c) Evaluate $\displaystyle\int_0^1 \left(te^t\,\mathbf{i} + \frac{2t}{1+t^2}\,\mathbf{j} + 3t^2e^{t^3}\,\mathbf{k}\right) dt$.

Solution P1

Calculus on each component

(a) Take the limit of each component separately. Each is a standard limit of the form $0/0$.

$$\lim_{t\to 0}\frac{\sin 3t}{t} = 3\lim_{t\to 0}\frac{\sin 3t}{3t} = 3, \qquad \lim_{t\to 0}\frac{1-\cos t}{t^2} = \frac{1}{2}, \qquad \lim_{t\to 0}\frac{e^{2t}-1}{t} = 2$$

The middle one follows from $1-\cos t = 2\sin^2(t/2)$, or from L'Hôpital applied twice. The last is the derivative of e^{2t} at $t=0$.

$$\boxed{\langle 3,\ \tfrac{1}{2},\ 2\rangle}$$

(b) Differentiate each component. The third needs the chain rule: $\dfrac{d}{dt}(1+t^2)^{-1} = -2t(1+t^2)^{-2}$.

$$\mathbf{r}'(t) = \left\langle (1+t)e^t,\ 2\cos 2t,\ -\frac{2t}{(1+t^2)^2}\right\rangle$$

Differentiate again, using the product rule on the third component:

$$\mathbf{r}''(t) = \left\langle (2+t)e^t,\ -4\sin 2t,\ -\frac{2}{(1+t^2)^2} + \frac{8t^2}{(1+t^2)^3}\right\rangle$$

At $t=0$: $(2)(1)=2$, $-4\sin 0 = 0$, and $-2+0=-2$.

$$\boxed{\mathbf{r}'(t) = \left\langle (1+t)e^t,\ 2\cos 2t,\ -\frac{2t}{(1+t^2)^2}\right\rangle, \qquad \mathbf{r}''(0) = \langle 2,\ 0,\ -2\rangle}$$

(c) Integrate each component from 0 to 1.

i: by parts, $\int te^t\,dt = (t-1)e^t$, so the value is $0-(-1)=1$.

j: $\displaystyle\int \frac{2t}{1+t^2}\,dt = \ln(1+t^2)$, so the value is $\ln 2$.

k: $\int 3t^2e^{t^3}\,dt = e^{t^3}$, so the value is $e-1$.

$$\boxed{\mathbf{i} + (\ln 2)\,\mathbf{j} + (e-1)\,\mathbf{k}}$$

Problem P2

(a) For $\mathbf{r}(t) = \langle e^t \cos t,\ e^t \sin t,\ e^t \rangle$, find the tangent vector, the unit tangent vector and the tangent line at $t = 0$, and the arc length from $t = 0$ to $t = \ln 3$.

(b) For $\mathbf{r}(t) = \langle 2t,\ t^2,\ \ln t \rangle$, find the arc length from $t = 1$ to $t = e$ and the unit tangent vector at $t = 1$.

Solution P2

Tangent vector, unit tangent, arc length

(a) The curve is a spiral that climbs a cone about the z-axis, since $x^2 + y^2 = z^2$ for every t. By the product rule,

$$\mathbf{r}'(t) = e^t\langle \cos t - \sin t,\ \sin t + \cos t,\ 1\rangle$$

At $t = 0$ the point is $\mathbf{r}(0) = (1, 0, 1)$ and the tangent vector is $\mathbf{r}'(0) = \langle 1, 1, 1\rangle$, with length $\sqrt{3}$. The unit tangent is

$$\mathbf{T}(0) = \frac{1}{\sqrt{3}}\langle 1,\ 1,\ 1\rangle$$

The tangent line through $(1, 0, 1)$ with direction $\langle 1, 1, 1\rangle$ is $x = 1 + s,\ y = s,\ z = 1 + s$.

For arc length we need $\|\mathbf{r}'(t)\|$ for all t. The cross terms cancel: $(\cos t - \sin t)^2 + (\sin t + \cos t)^2 = 2$, so

$$\|\mathbf{r}'(t)\| = e^t\sqrt{2 + 1} = \sqrt{3}\, e^t$$

$$s = \int_0^{\ln 3} \sqrt{3}\, e^t\, dt = \sqrt{3}\,(3 - 1) = 2\sqrt{3}$$

$$\boxed{\mathbf{T}(0) = \tfrac{1}{\sqrt{3}}\langle 1, 1, 1\rangle, \qquad \text{tangent line } (x, y, z) = (1 + s,\ s,\ 1 + s), \qquad s = 2\sqrt{3}}$$

(b) $\mathbf{r}'(t) = \langle 2,\ 2t,\ 1/t\rangle$, so

$$\|\mathbf{r}'(t)\|^2 = 4 + 4t^2 + \frac{1}{t^2} = \Big(2t + \frac{1}{t}\Big)^2$$

The expression under the root is a perfect square, which is the reason for the choice of curve. For $t > 0$ the root is $2t + 1/t$:

$$s = \int_1^e \Big(2t + \frac{1}{t}\Big)dt = \Big[t^2 + \ln t\Big]_1^e = (e^2 + 1) - (1 + 0) = e^2$$

At $t = 1$: $\mathbf{r}'(1) = \langle 2, 2, 1\rangle$, with length $2t + 1/t = 3$.

$$\boxed{s = e^2, \qquad \mathbf{T}(1) = \tfrac{1}{3}\langle 2,\ 2,\ 1\rangle}$$

Problem P3

Let $f(x, y, z) = x^2y - yz^2 + xz$. Find the directional derivative of f at $P(1, -2, 3)$ in the direction of the point $Q(3, 0, 2)$. Then give the largest rate of change of f at P and the unit vector in which it occurs.

Solution P3

Gradient, unit vector, dot product

Partial derivatives:

$$f_x = 2xy + z, \qquad f_y = x^2 - z^2, \qquad f_z = -2yz + x$$

At $P(1, -2, 3)$: $f_x = -4 + 3 = -1$, $f_y = 1 - 9 = -8$, $f_z = 12 + 1 = 13$, so

$$\nabla f(P) = \langle -1,\ -8,\ 13 \rangle$$

The direction from P toward Q is $\overrightarrow{PQ} = \langle 2, 2, -1 \rangle$. It has length $\sqrt{4 + 4 + 1} = 3$, so the **unit** vector is $\mathbf{u} = \frac{1}{3}\langle 2, 2, -1 \rangle$. Using the unnormalized vector would give a result three times too large.

$$D_{\mathbf{u}} f(P) = \nabla f \cdot \mathbf{u} = \frac{(-1)(2) + (-8)(2) + (13)(-1)}{3} = -\frac{31}{3}$$

Negative, so f decreases as you move from P toward Q.

The largest rate of change is $\|\nabla f(P)\| = \sqrt{1 + 64 + 169} = \sqrt{234} = 3\sqrt{26}$, in the direction

$$\mathbf{u}_{\max} = \frac{\langle -1,\ -8,\ 13 \rangle}{3\sqrt{26}}$$

$$\boxed{D_{\mathbf{u}} f(P) = -\tfrac{31}{3} \text{ with } \mathbf{u} = \tfrac{1}{3}\langle 2, 2, -1 \rangle; \qquad \max = 3\sqrt{26} \text{ along } \tfrac{1}{3\sqrt{26}}\langle -1, -8, 13 \rangle}$$

As a sanity check, $-31/3 \approx -10.33$ has magnitude below $3\sqrt{26} \approx 15.30$, as it must.

Problem P4

Let $f(x, y) = \ln(x^2 + y^2)$ and $P = (1, 2)$.

(a) Find the rate of change of f at P in the direction of $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$.

(b) In which direction does f increase most rapidly at P, and what is that rate?

(c) In which directions is the rate of change zero at P?

Solution P4

Normalize, then use the gradient

$$f_x = \frac{2x}{x^2+y^2}, \qquad f_y = \frac{2y}{x^2+y^2}, \qquad \nabla f(1,2) = \Big\langle \frac{2}{5},\ \frac{4}{5} \Big\rangle$$

(a) $\mathbf{a} = \langle 3, -4\rangle$ is not a unit vector: $\|\mathbf{a}\| = 5$. So $\mathbf{u} = \langle \frac{3}{5}, -\frac{4}{5}\rangle$ and

$$D_{\mathbf{u}} f(1,2) = \frac{2}{5}\cdot\frac{3}{5} + \frac{4}{5}\cdot\Big(-\frac{4}{5}\Big) = \frac{6-16}{25} = -\frac{2}{5}$$

(b) The maximum rate is $\|\nabla f\| = \dfrac{1}{5}\sqrt{4+16} = \dfrac{2\sqrt{5}}{5} = \dfrac{2}{\sqrt{5}}$, in the direction

$$\mathbf{u}_{\max} = \frac{\langle 2/5,\ 4/5\rangle}{2/\sqrt{5}} = \frac{1}{\sqrt{5}}\langle 1,\ 2\rangle$$

This points straight away from the origin. That is expected: f depends only on the distance from the origin, and grows with it.

(c) The rate is zero when $\mathbf{u} \perp \nabla f$, that is when $\mathbf{u}\cdot\langle 1,2\rangle = 0$. The two unit vectors are

$$\mathbf{u} = \pm\frac{1}{\sqrt{5}}\langle 2,\ -1\rangle$$

These are tangent to the level curve $x^2+y^2 = 5$ through P, which is a circle about the origin.

(a) $-\frac{2}{5}$; (b) $\frac{2}{\sqrt{5}}$ along $\frac{1}{\sqrt{5}}\langle 1,2\rangle$; (c) $\pm\frac{1}{\sqrt{5}}\langle 2,-1\rangle$

Problem P5

Find the equation of the tangent plane and the parametric equations of the normal line to the surface $x^2 + 2y^2 + 3z^2 = 21$ at $(1, 2, 2)$. Then find where the normal line crosses the xy-plane.

Solution P5

Level surface, gradient is the normal

First confirm the point is on the surface: $1 + 8 + 12 = 21$. Good. Take $F = x^2 + 2y^2 + 3z^2$, so $c = 21$ and

$$\nabla F = \langle 2x,\ 4y,\ 6z\rangle, \qquad \nabla F(1,2,2) = \langle 2,\ 8,\ 12\rangle = 2\langle 1,4,6\rangle$$

A multiple of the normal is just as good as the normal. Using $\langle 1,4,6\rangle$:

$$1\,(x-1) + 4\,(y-2) + 6\,(z-2) = 0 \implies x + 4y + 6z = 21$$

The plane is $x + 4y + 6z = 21$. The normal line passes through $(1,2,2)$ with direction $\langle 1,4,6\rangle$:

$$x = 1 + t, \qquad y = 2 + 4t, \qquad z = 2 + 6t$$

It meets the xy-plane where $z = 0$: $2 + 6t = 0$, so $t = -\frac{1}{3}$. Then $x = \frac{2}{3}$ and $y = 2 - \frac{4}{3} = \frac{2}{3}$.

$$\boxed{x + 4y + 6z = 21; \qquad (x,y,z) = (1 + t,\ 2 + 4t,\ 2 + 6t); \qquad \text{crosses at } (\tfrac{2}{3}, \tfrac{2}{3}, 0)}$$

Problem P6

Find the tangent plane and the normal line to the graph of $z = x^2 e^y$ at the point where $x = 1$ and $y = \ln 3$.

Solution P6

A graph is a level surface of $f(x,y)-z$

At $(1, \ln 3)$: $z = 1 \cdot e^{\ln 3} = 3$, so the point is $(1,\ \ln 3,\ 3)$. Write the graph as $F(x,y,z) = x^2 e^y - z = 0$. Then

$$\nabla F = \langle 2xe^y,\ x^2 e^y,\ -1\rangle, \qquad \nabla F(1, \ln 3,\ 3) = \langle 6,\ 3,\ -1\rangle$$

Tangent plane:

$$6(x-1) + 3(y - \ln 3) - (z-3) = 0 \implies 6x + 3y - z = 3 + 3\ln 3$$

Equivalently $z = 3 + 6(x-1) + 3(y - \ln 3)$, the usual form $z - z_0 = f_x(x-x_0) + f_y(y-y_0)$. Normal line:

$$x = 1 + 6t, \qquad y = \ln 3 + 3t, \qquad z = 3 - t$$

$$\boxed{6x + 3y - z = 3 + 3\ln 3; \qquad (x,y,z) = (1+6t,\ \ln 3 + 3t,\ 3 - t)}$$

Problem P7

(a) At what point of the surface $z = x^2 - 2x + y^2 + 4y$ is the tangent plane horizontal? Give the plane.

(b) At what points of the ellipsoid $x^2 + 2y^2 + z^2 = 16$ is the tangent plane parallel to the plane $x + 2y + z = 5$? Give the tangent planes.

Solution P7

The normal must have the required direction

(a) A horizontal plane has a vertical normal, so $\langle f_x, f_y, -1\rangle$ must be vertical, meaning $f_x = f_y = 0$.

$$f_x = 2x - 2 = 0 \Rightarrow x = 1, \qquad f_y = 2y + 4 = 0 \Rightarrow y = -2$$

Then $z = 1 - 2 + 4 - 8 = -5$. The point is $(1, -2, -5)$ and the plane is $z = -5$. (Completing the square gives $z = (x-1)^2 + (y+2)^2 - 5$, a bowl opening upward with its lowest point there.)

(b) With $F = x^2 + 2y^2 + z^2$, the normal is $\nabla F = \langle 2x, 4y, 2z\rangle$. It must be a multiple of the normal $\langle 1, 2, 1\rangle$ of the given plane:

$$2x = \lambda, \quad 4y = 2\lambda, \quad 2z = \lambda \implies x = y = z = \frac{\lambda}{2} \equiv s$$

Put this on the ellipsoid: $s^2 + 2s^2 + s^2 = 4s^2 = 16$, so $s = \pm 2$. The points are $(2, 2, 2)$ and $(-2, -2, -2)$.

At $(2, 2, 2)$ the normal is $\langle 4, 8, 4\rangle \parallel \langle 1, 2, 1\rangle$, so $(x-2) + 2(y-2) + (z-2) = 0$, which is $x + 2y + z = 8$.

At $(-2, -2, -2)$ the same normal direction gives $(x+2) + 2(y+2) + (z+2) = 0$, which is $x + 2y + z = -8$.

(a) $(1, -2, -5)$, $z = -5$; (b) $(2, 2, 2)$ with $x + 2y + z = 8$, $(-2, -2, -2)$ with $x + 2y + z = -8$

The two planes sit on opposite sides of the ellipsoid, equally far from the origin, as the symmetry of the surface requires.

Problem P8

The sphere $x^2+y^2+z^2=6$ and the paraboloid $z=x^2+y^2$ intersect in a circle. Find the angle between the two surfaces at the point $(1,1,2)$ of that circle.

Solution P8

Angle between the normals

First check the point lies on both. Sphere: $1 + 1 + 4 = 6$. Paraboloid: $z = 2 = 1 + 1$. Both hold. (Substituting $x^2 + y^2 = z$ into the sphere gives $z + z^2 = 6$, so $z = 2$ since $z = x^2 + y^2 \geq 0$ rules out $z = -3$. The intersection is the circle $x^2 + y^2 = 2$ at height $z = 2$.)

Normals:

$$F = x^2 + y^2 + z^2 : \quad \nabla F = \langle 2x, 2y, 2z \rangle, \qquad \nabla F(1, 1, 2) = \langle 2, 2, 4 \rangle$$

$$G = x^2 + y^2 - z : \quad \nabla G = \langle 2x, 2y, -1 \rangle, \qquad \nabla G(1, 1, 2) = \langle 2, 2, -1 \rangle$$

$$\nabla F \cdot \nabla G = 4 + 4 - 4 = 4, \qquad \|\nabla F\| = \sqrt{24} = 2\sqrt{6}, \qquad \|\nabla G\| = 3$$

$$\cos\theta = \frac{4}{2\sqrt{6} \cdot 3} = \frac{2}{3\sqrt{6}} = \frac{\sqrt{6}}{9}$$

$$\boxed{\theta = \arccos\Big(\frac{\sqrt{6}}{9}\Big) \approx 74.2^\circ}$$

The cosine is positive, so θ is the acute angle between the surfaces. By the rotational symmetry about the z-axis, every point of the circle gives the same angle. Had the dot product come out negative, the angle between the surfaces would still be quoted as the acute one, $\arccos|\cos\theta|$.